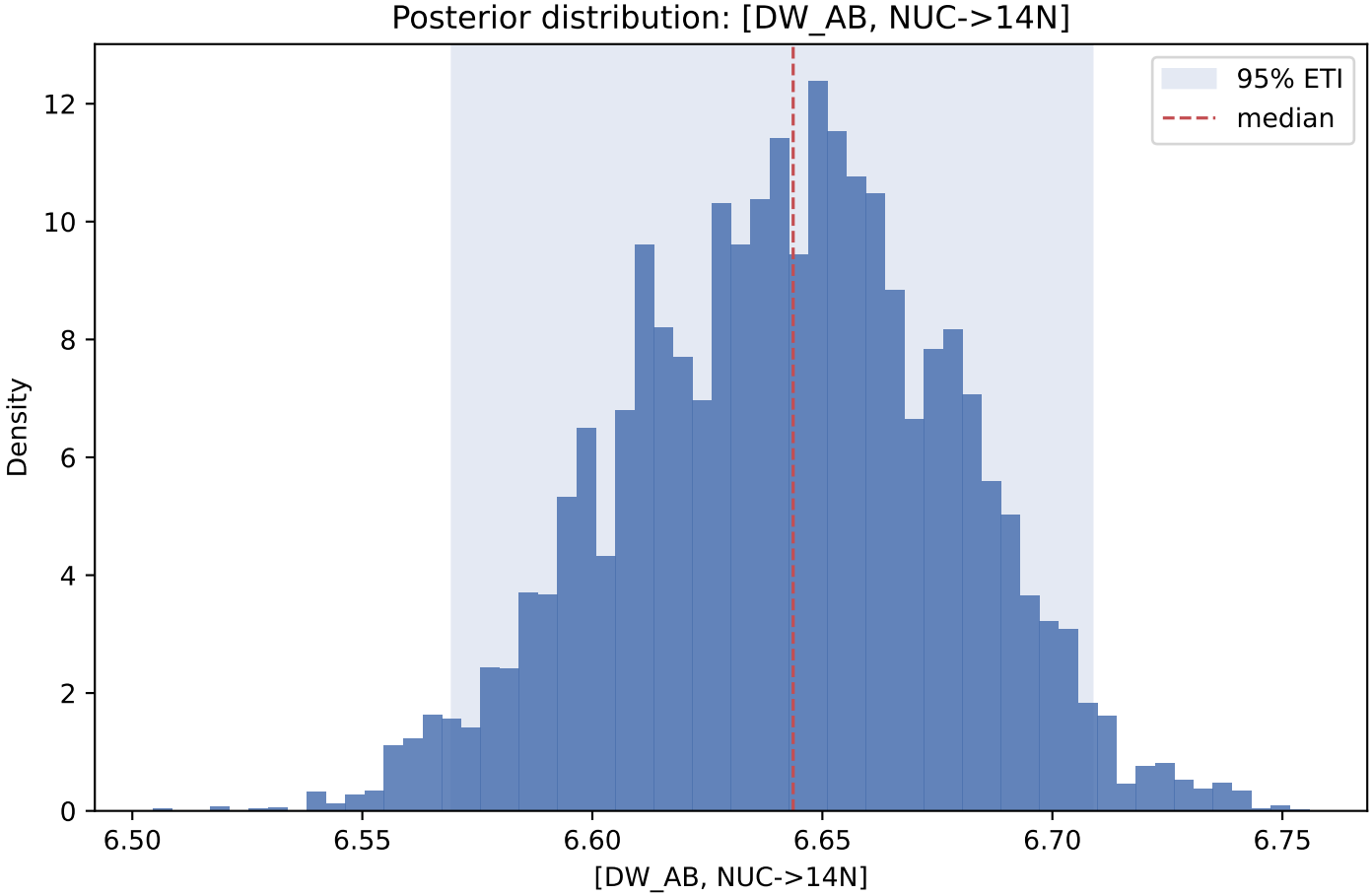
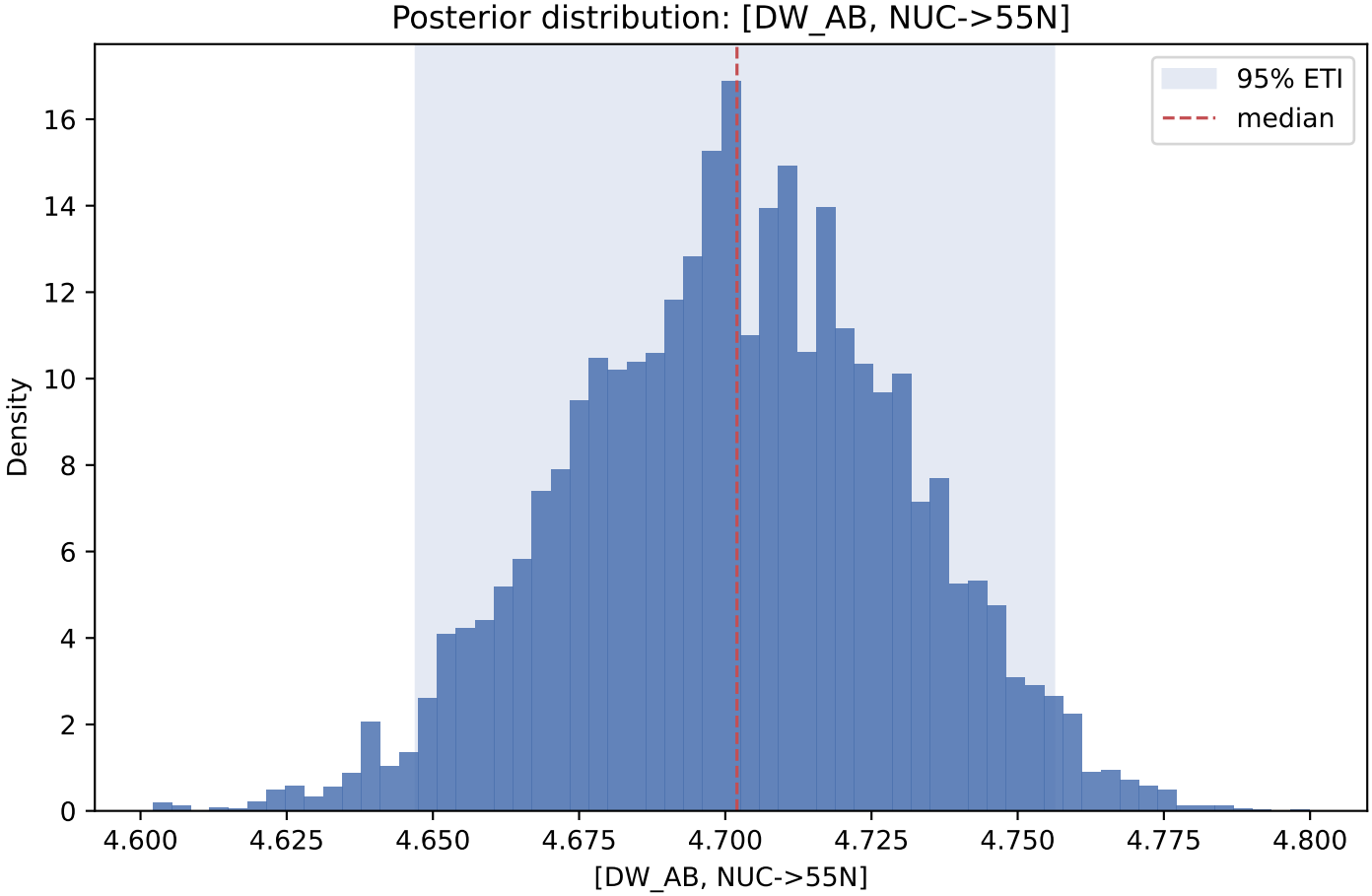
MCMC Diagnostics

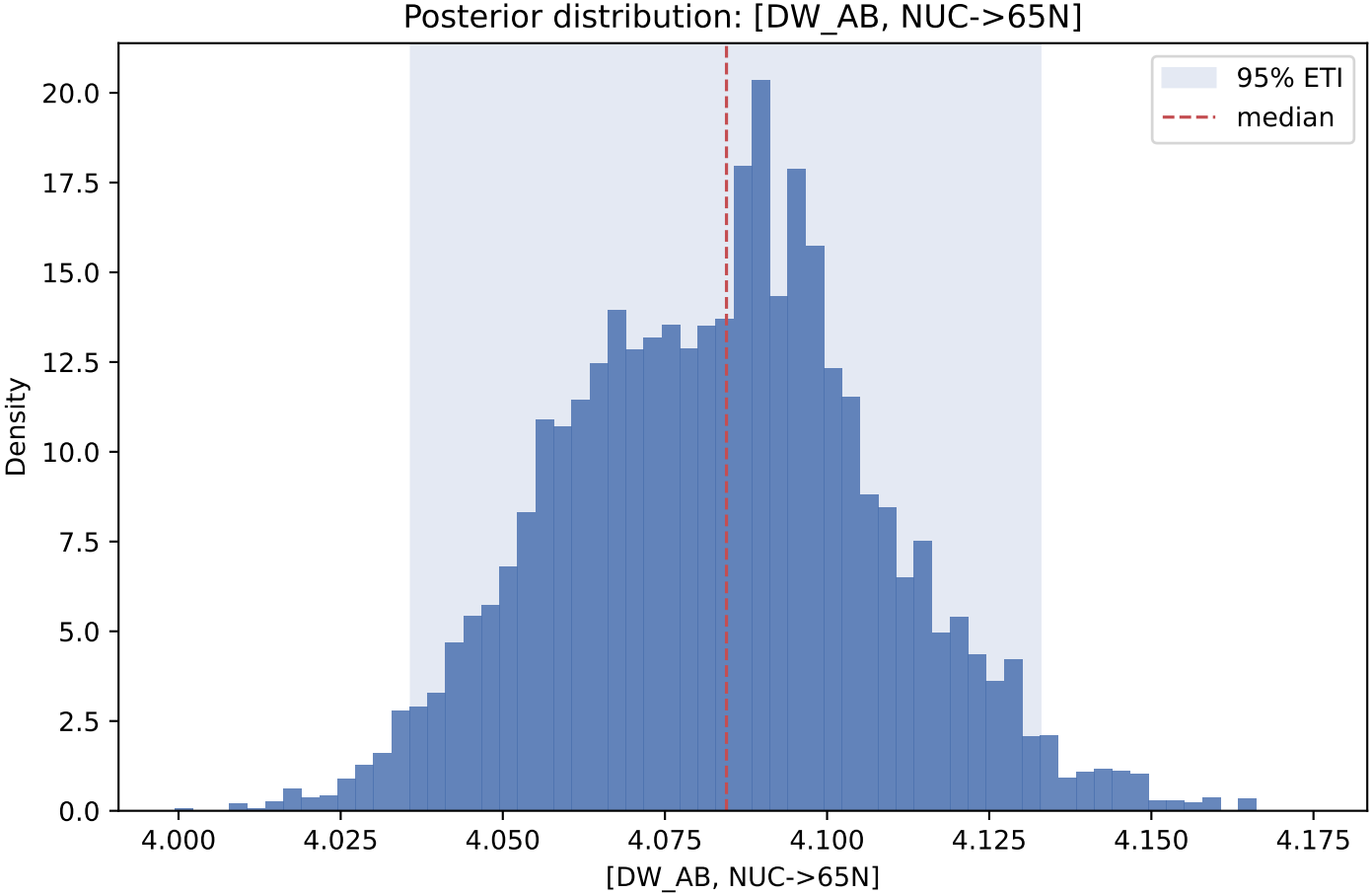
```
Sampler: emcee via ChemEx direct EnsembleSampler
Parameters: 11
Steps: 500
Walkers: 32
Requested burn: "auto"
Discarded steps: 83
Thin: 1
Retained steps: 417
Retained samples: 13344
Acceptance fraction: mean=0.40194, min=0.346, max=0.44
Autocorrelation status: unreliable short chain
Max autocorrelation time: 41.266
Recommended steps (50 tau): 2064
Recommended steps (100 tau): 4127
Autocorrelation warning: chain shorter than 50 times the autocorrelation time; tentative estimate rep
Burn-in warning: autocorrelation time estimate is unreliable; tentative automatic burn-in was applied

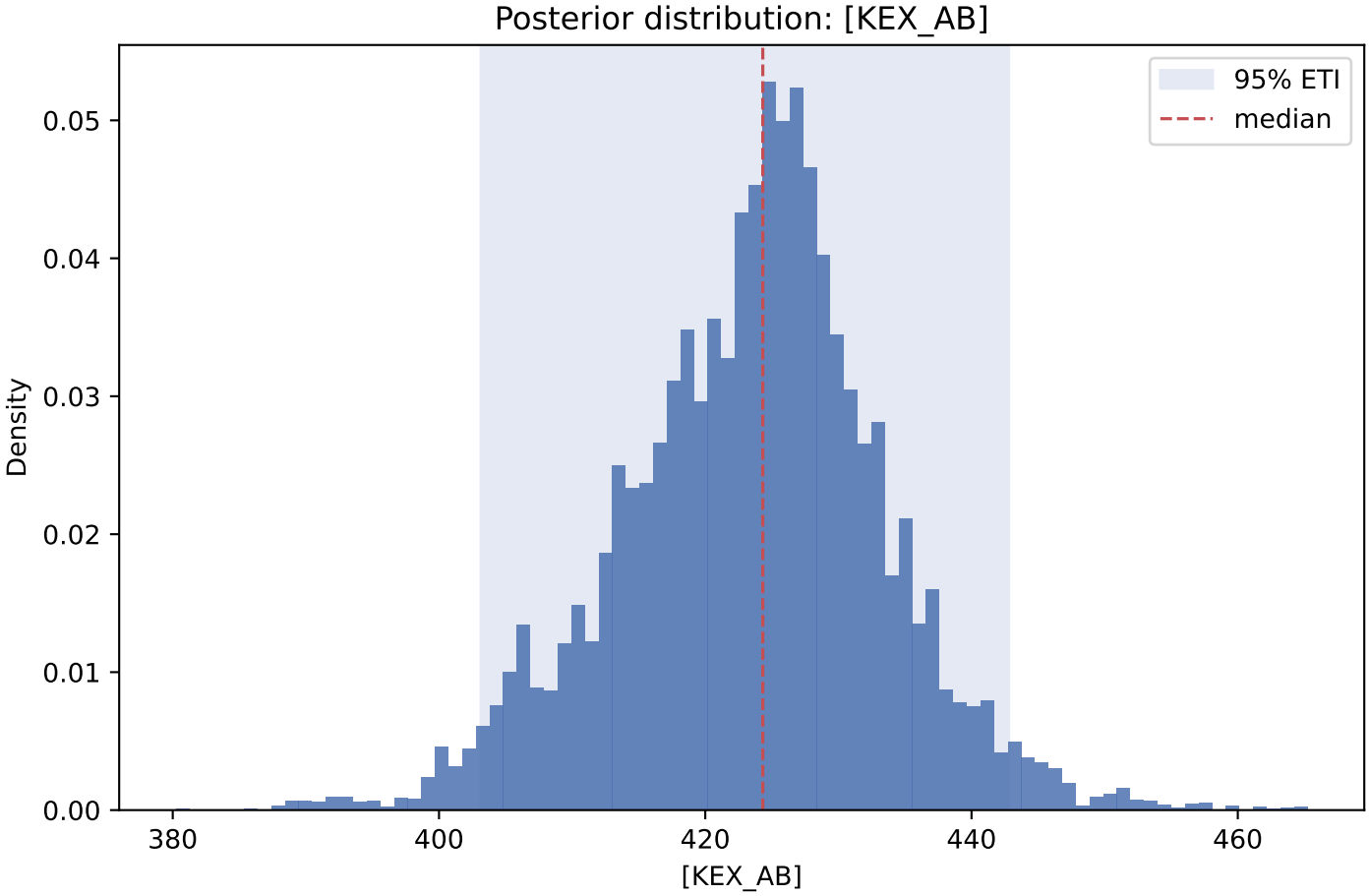
Posterior summaries:
[DW_AB, NUC->14N]: median=6.6437, 95% ETI=[6.5695, 6.7087], ESS=n/a
[DW_AB, NUC->55N]: median=4.702, 95% ETI=[4.6471, 4.7562], ESS=n/a
[DW_AB, NUC->65N]: median=4.0845, 95% ETI=[4.0359, 4.1328], ESS=n/a
[KEX_AB]: median=424.32, 95% ETI=[403.14, 442.8], ESS=n/a
[PB]: median=0.003479, 95% ETI=[0.0034069, 0.0035498], ESS=n/a
[R1_A, NUC->14N, B0->600.3MHZ]: median=1.9661, 95% ETI=[1.9509, 1.9813], ESS=n/a
[R1_A, NUC->55N, B0->600.3MHZ]: median=1.5975, 95% ETI=[1.5865, 1.6079], ESS=n/a
[R1_A, NUC->65N, B0->600.3MHZ]: median=1.6072, 95% ETI=[1.5982, 1.6162], ESS=n/a
[R2_A, NUC->14N, B0->600.3MHZ]: median=10.638, 95% ETI=[10.473, 10.84], ESS=n/a
[R2_A, NUC->55N, B0->600.3MHZ]: median=9.8827, 95% ETI=[9.7636, 10.005], ESS=n/a
[R2_A, NUC->65N, B0->600.3MHZ]: median=9.3018, 95% ETI=[9.2017, 9.3973], ESS=n/a

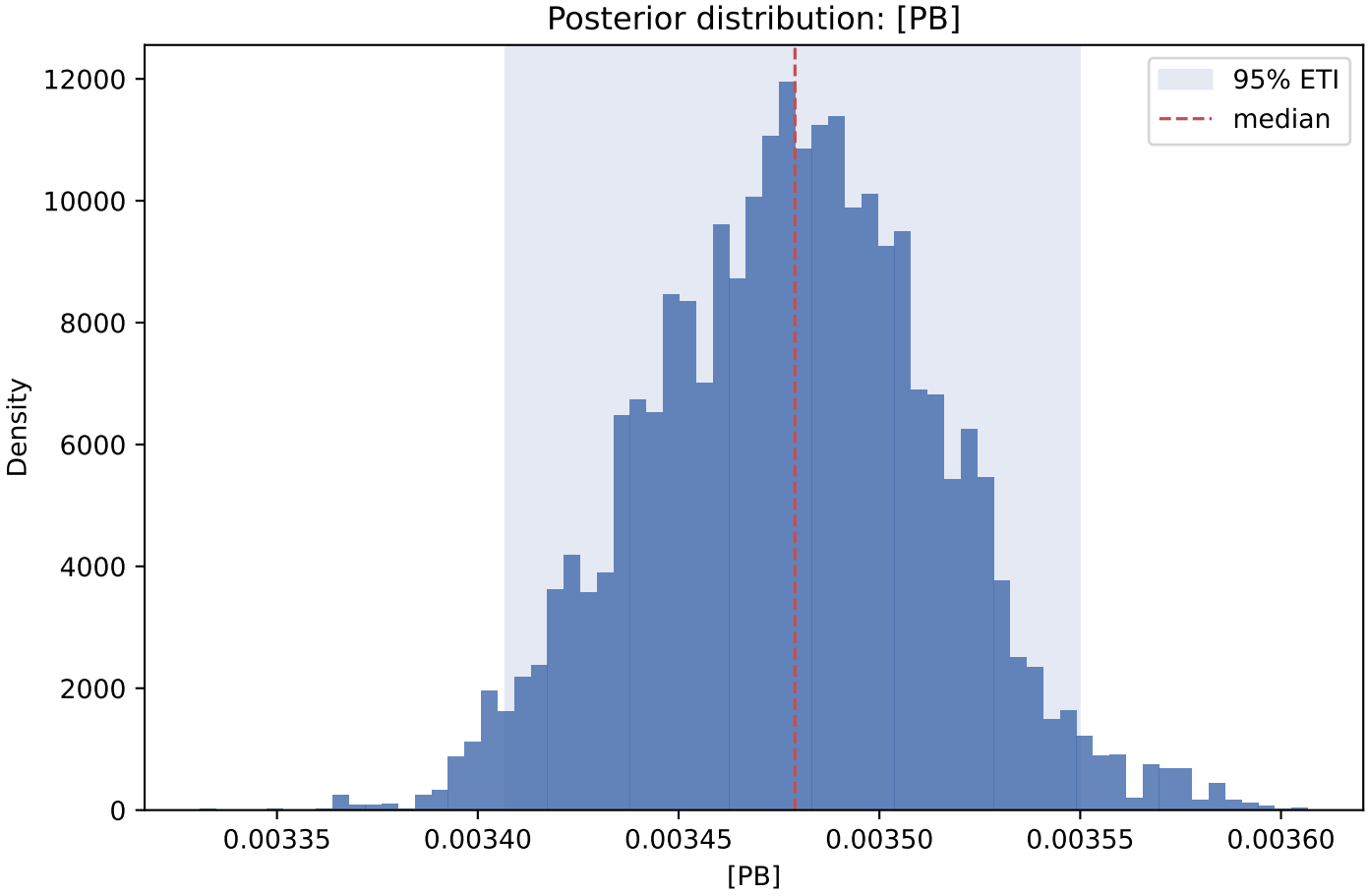
Correlation threshold: |r| >= 0.5
No parameter pairs exceeded the correlation threshold.
```

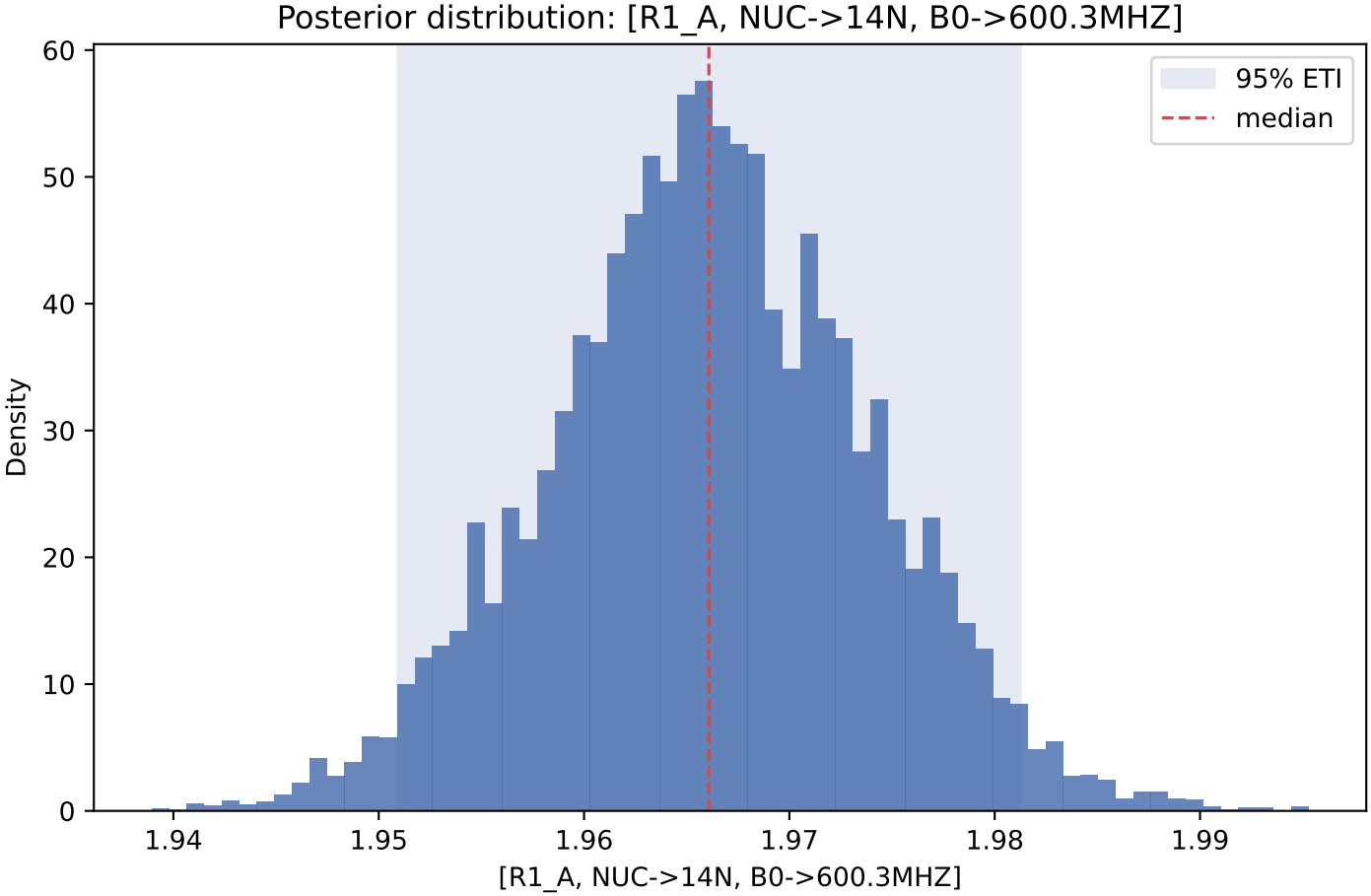

Posterior distribution: [DW_AB, NUC->14N]
95% ETI
median
Density
0
2
4
6
8
10
12
6.50
6.55
6.60
6.65
6.70
6.75
[DW_AB, NUC->14N]

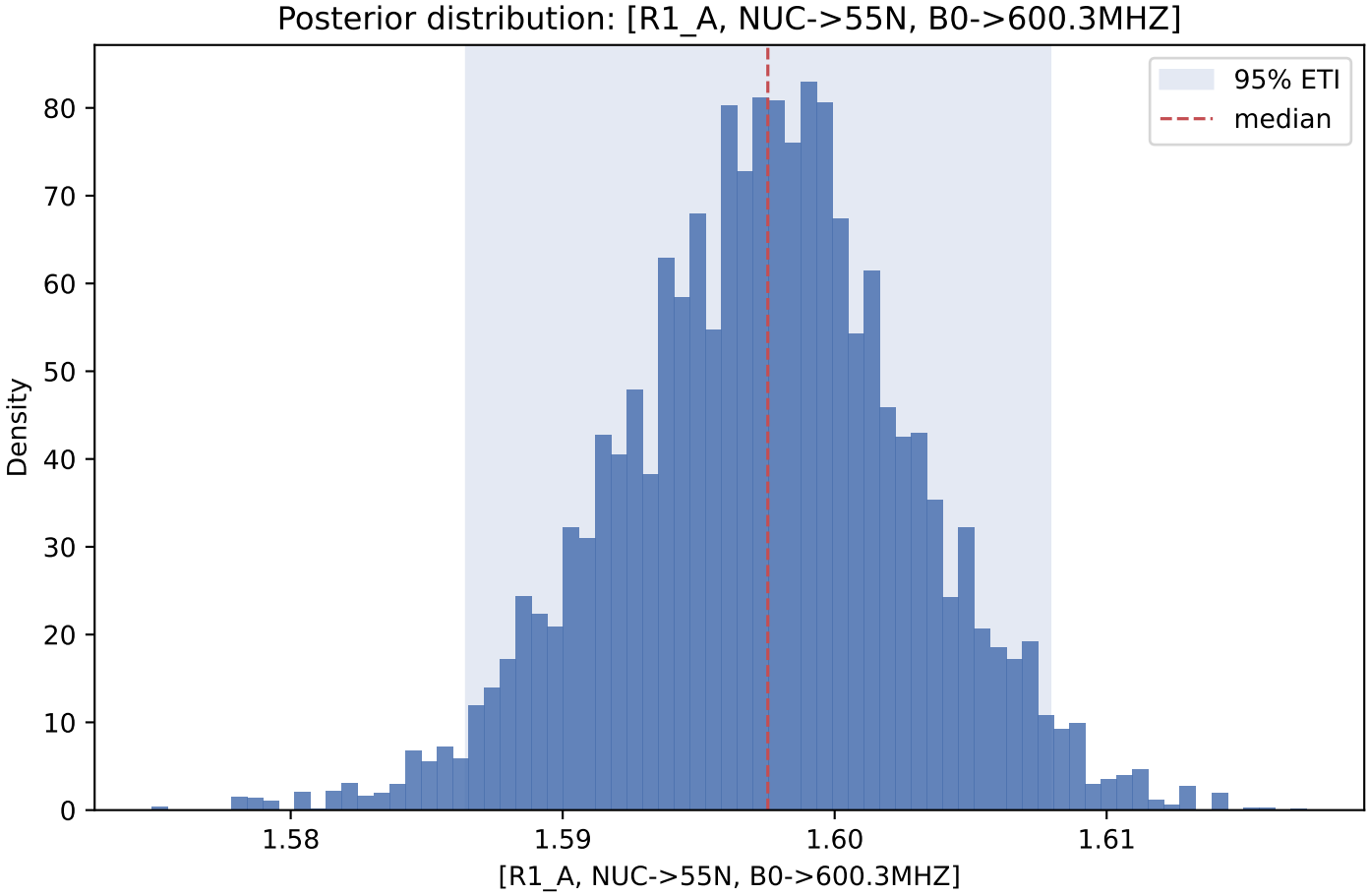

Posterior distribution: [DW_AB, NUC->55N]
95% ETI
median
Density
0
2
4
6
8
10
12
14
16
4.600
4.625
4.650
4.675
4.700
4.725
4.750
4.775
4.800
[DW_AB, NUC->55N]

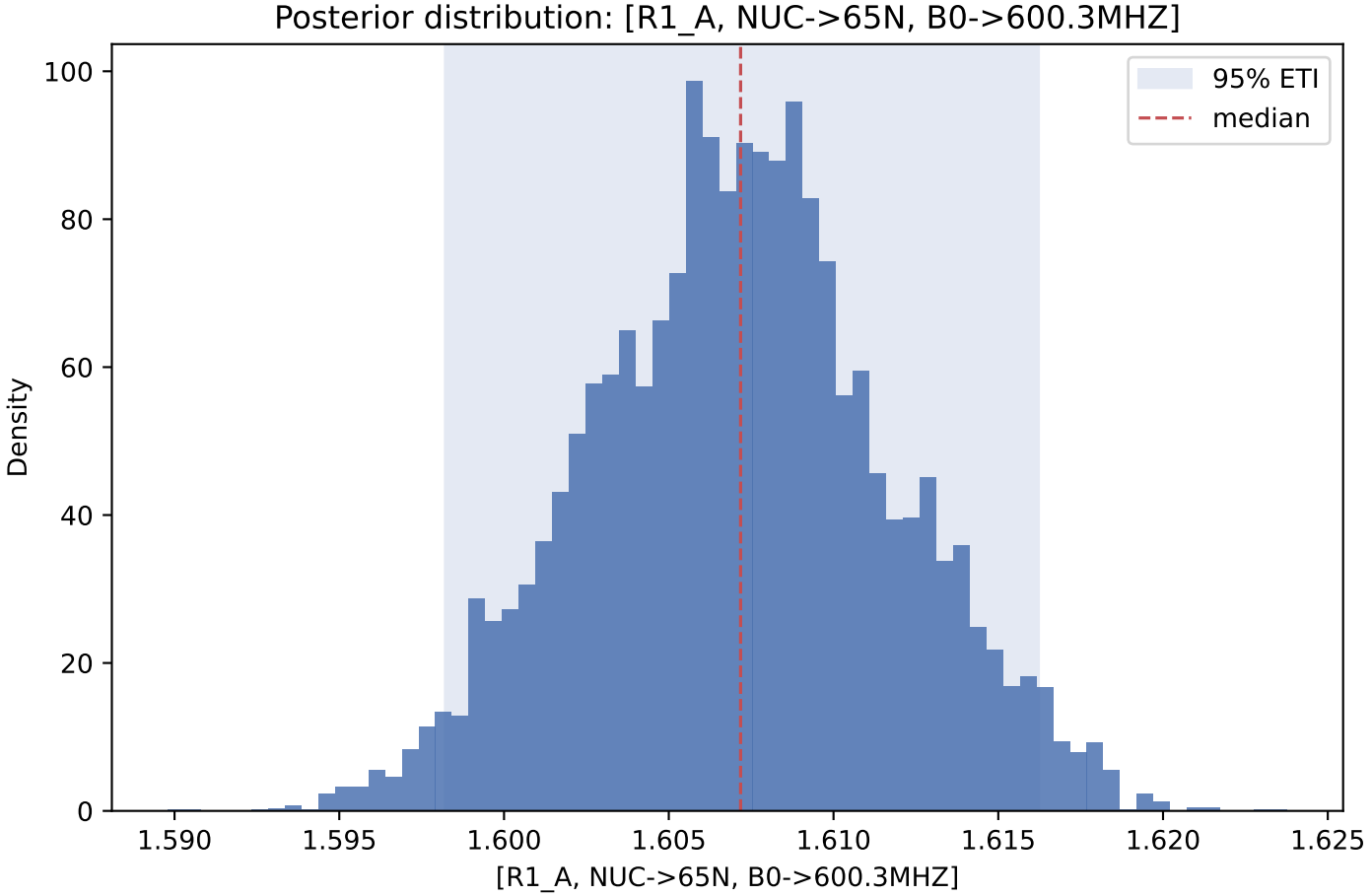

Posterior distribution: [DW_AB, NUC->65N]
95% ETI
median
Density
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
4.000
4.025
4.050
4.075
4.100
4.125
4.150
4.175
[DW_AB, NUC->65N]

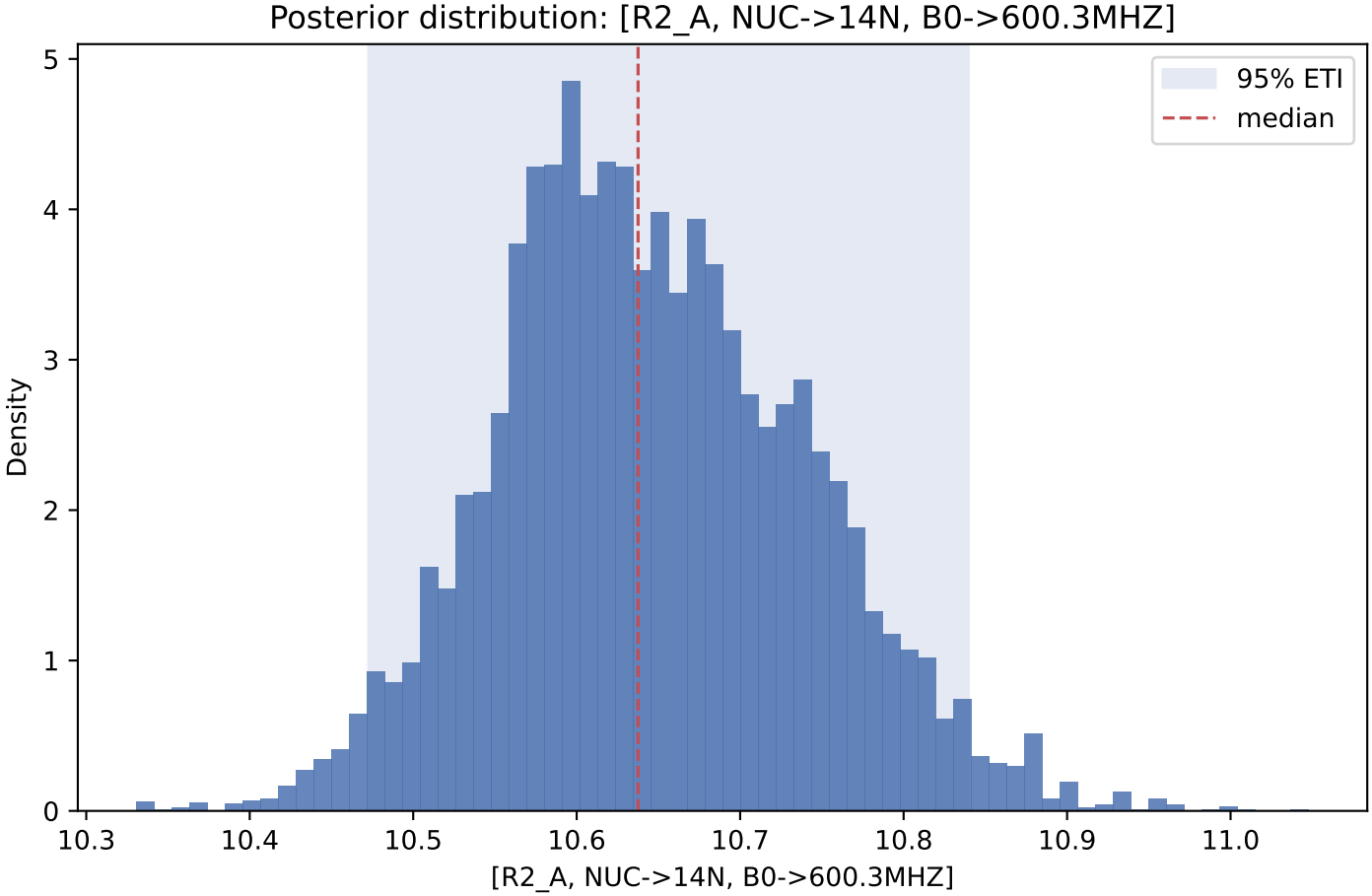

Posterior distribution: [KEX_AB]
95% ETI
median
Density
0.00
0.01
0.02
0.03
0.04
0.05
380
400
420
440
460
[KEX_AB]

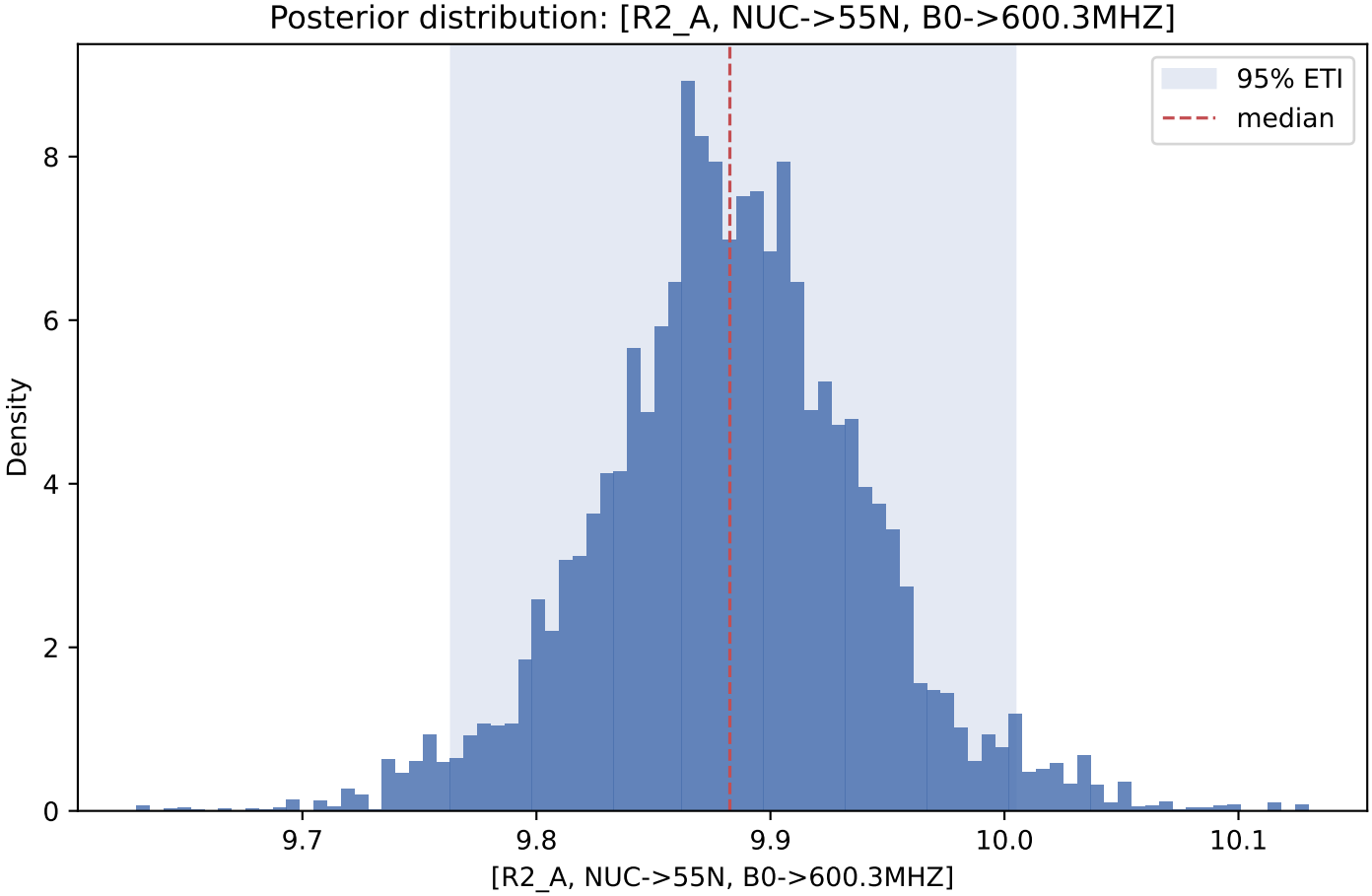

Posterior distribution: [PB]
95% ETI
median
Density
12000
10000
8000
6000
4000
2000
0
0.00335
0.00340
0.00345
0.00350
0.00355
0.00360
[PB]

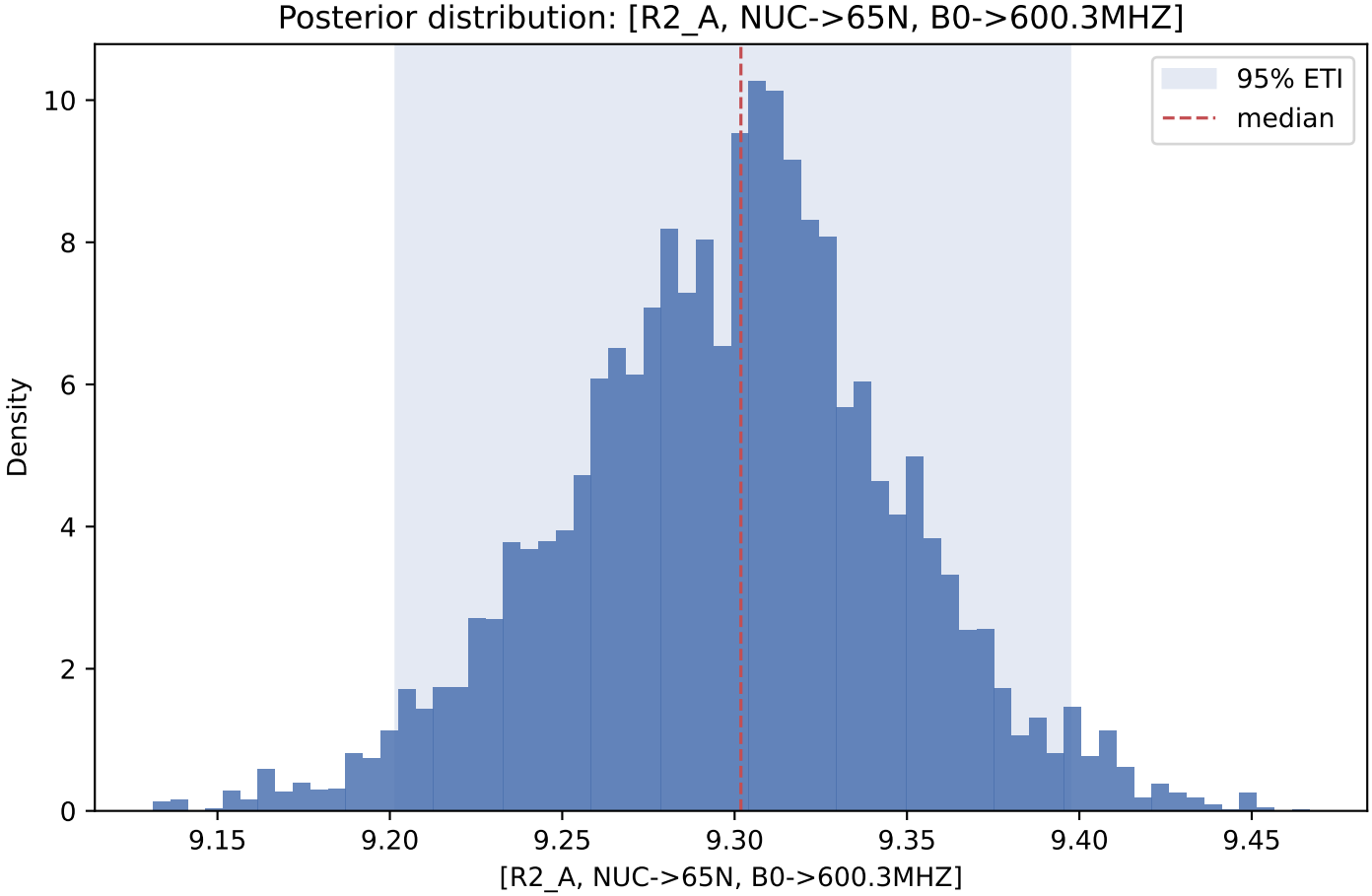

Posterior distribution: [R1_A, NUC->14N, B0->600.3MHZ]
95% ETI
median
Density
0
10
20
30
40
50
60
1.94
1.95
1.96
1.97
1.98
1.99
[R1_A, NUC->14N, B0->600.3MHZ]

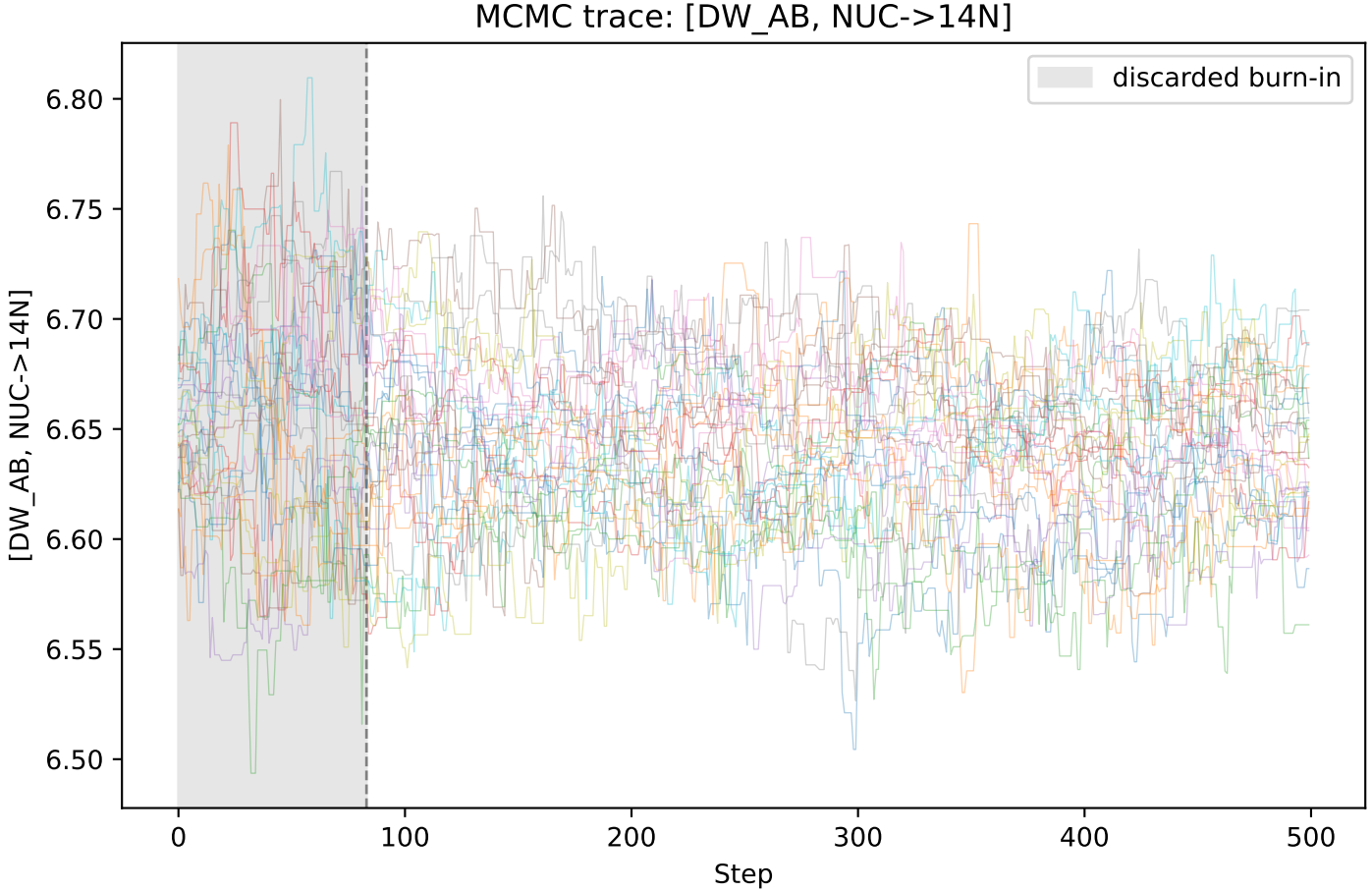

Posterior distribution: [R1_A, NUC->55N, B0->600.3MHZ]
95% ETI
median
Density
0
10
20
30
40
50
60
70
80
1.58
1.59
1.60
1.61
[R1_A, NUC->55N, B0->600.3MHZ]

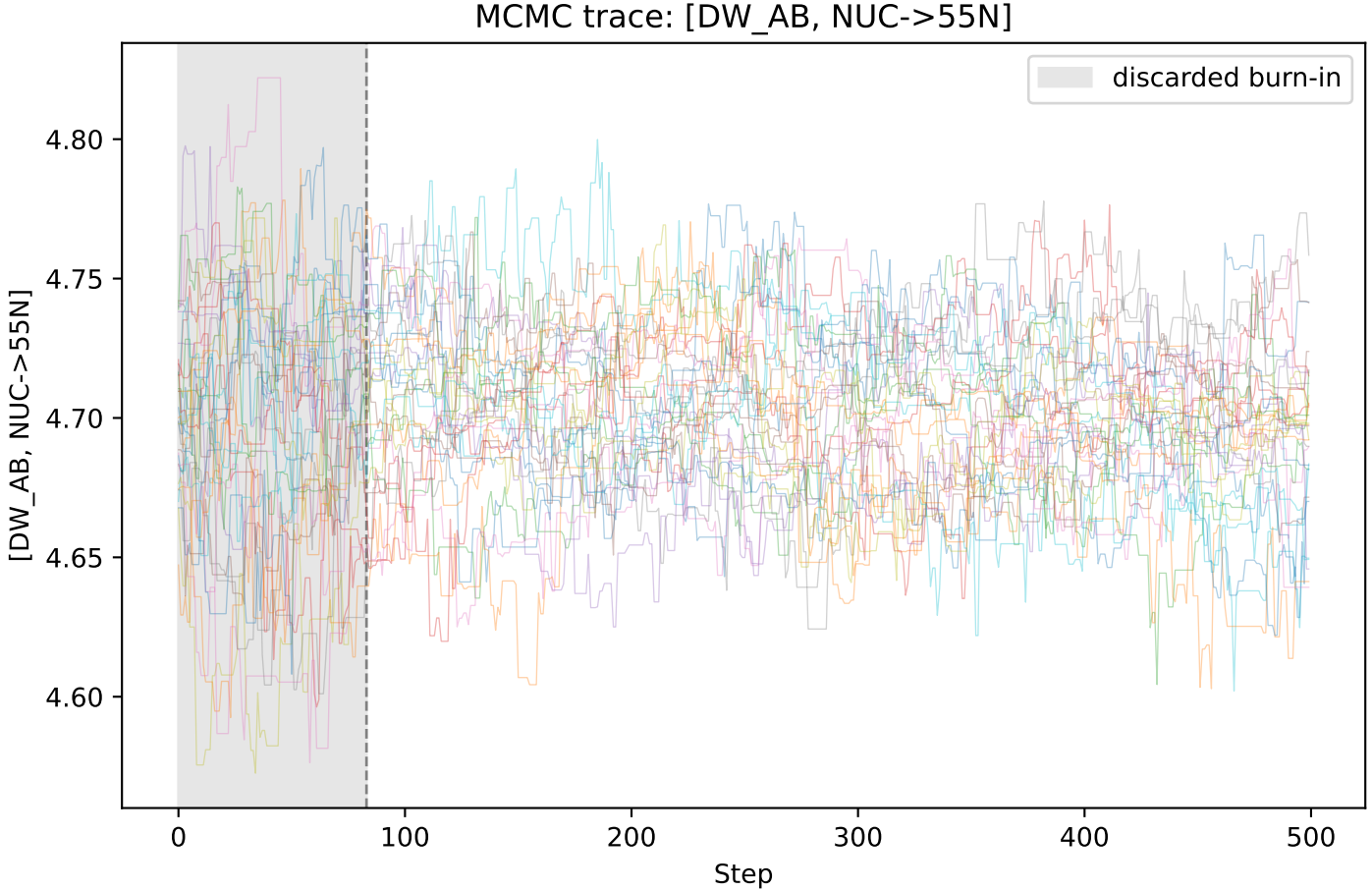

Posterior distribution: [R1_A, NUC->65N, B0->600.3MHZ]
95% ETI
median
Density
0
20
40
60
80
100
1.590
1.595
1.600
1.605
1.610
1.615
1.620
1.625
[R1_A, NUC->65N, B0->600.3MHZ]

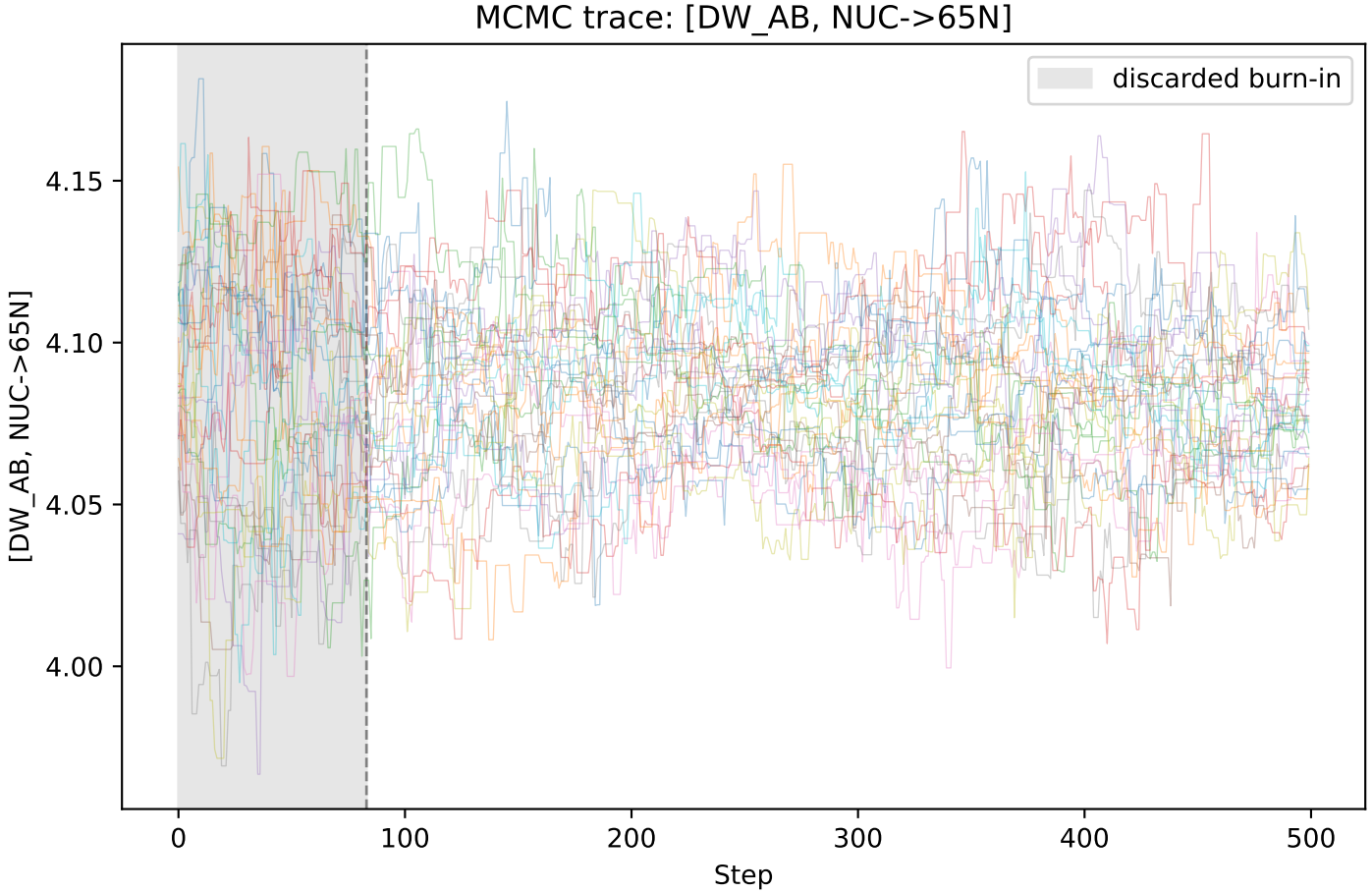

Posterior distribution: [R2_A, NUC->14N, B0->600.3MHZ]
95% ETI
median
Density
0
1
2
3
4
5
10.3
10.4
10.5
10.6
10.7
10.8
10.9
11.0
[R2_A, NUC->14N, B0->600.3MHZ]

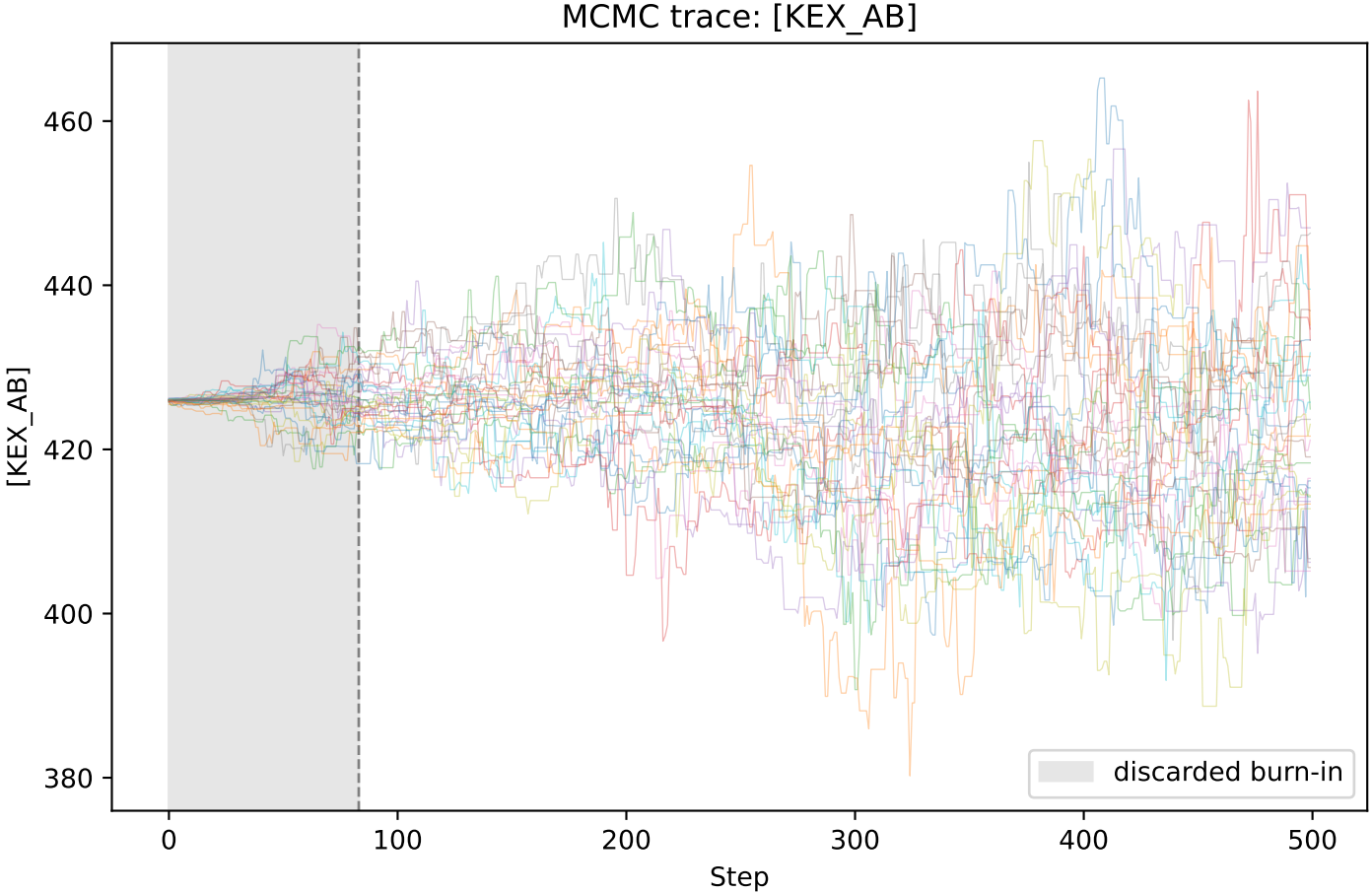

Posterior distribution: [R2_A, NUC->55N, B0->600.3MHZ]
95% ETI
median
Density
0
2
4
6
8
9.7
9.8
9.9
10.0
10.1
[R2_A, NUC->55N, B0->600.3MHZ]

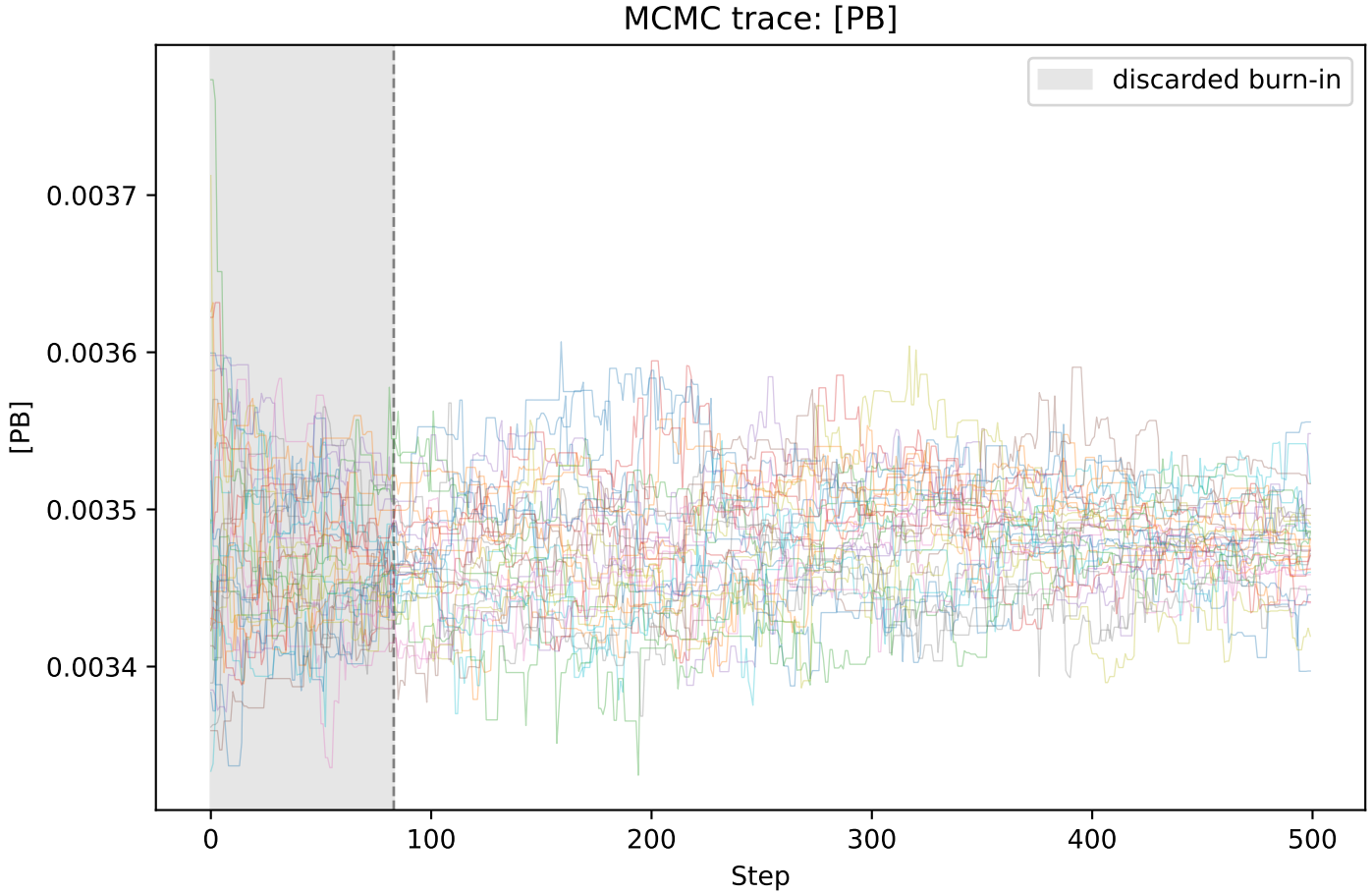

Posterior distribution: [R2_A, NUC->65N, B0->600.3MHZ]
95% ETI
median
Density
0
2
4
6
8
10
9.15
9.20
9.25
9.30
9.35
9.40
9.45
[R2_A, NUC->65N, B0->600.3MHZ]

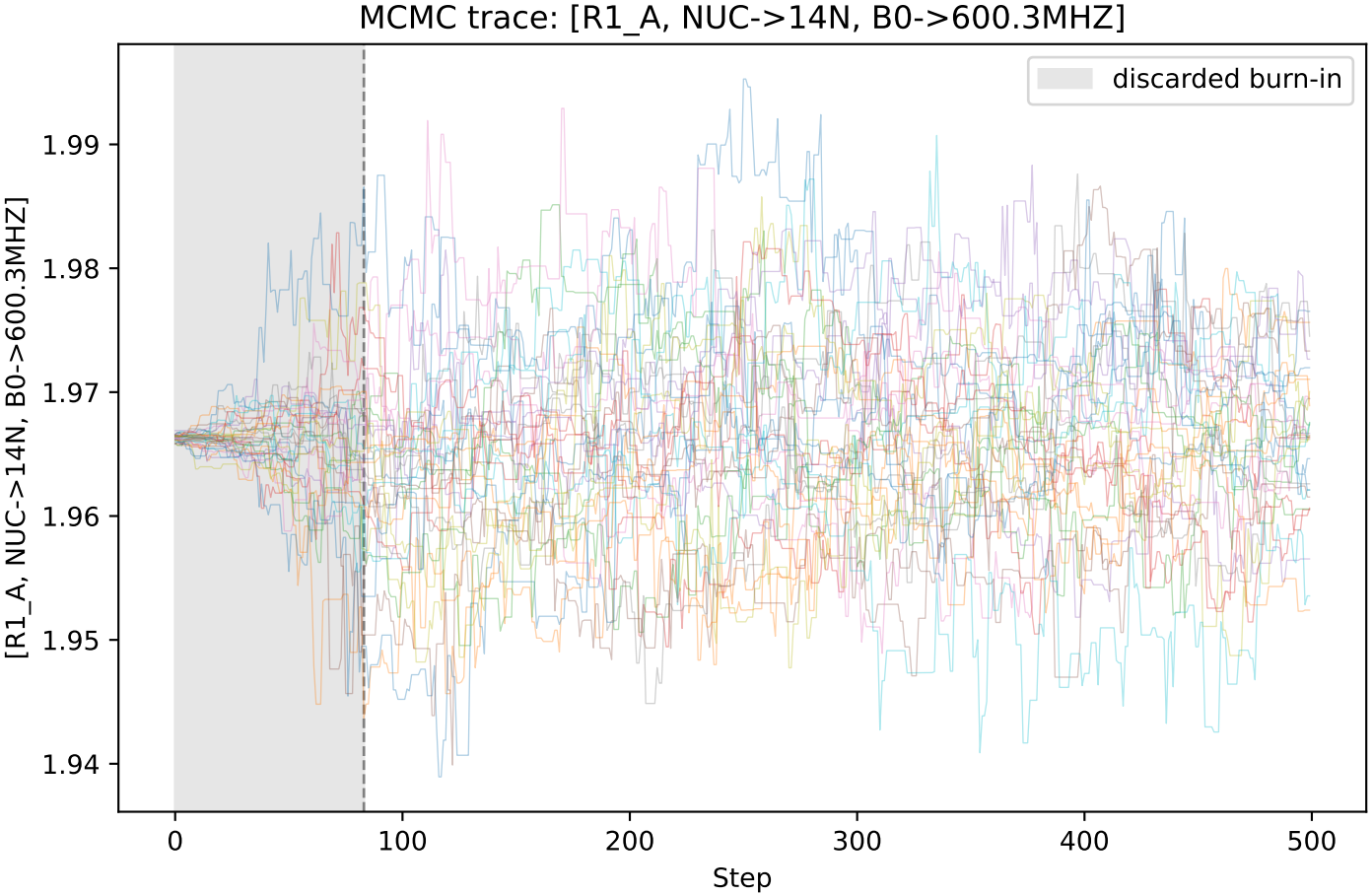

MCMC trace: [DW_AB, NUC->14N]
discarded burn-in
[DW_AB, NUC->14N]
6.80
6.75
6.70
6.65
6.60
6.55
6.50
0
100
200
300
400
500
Step

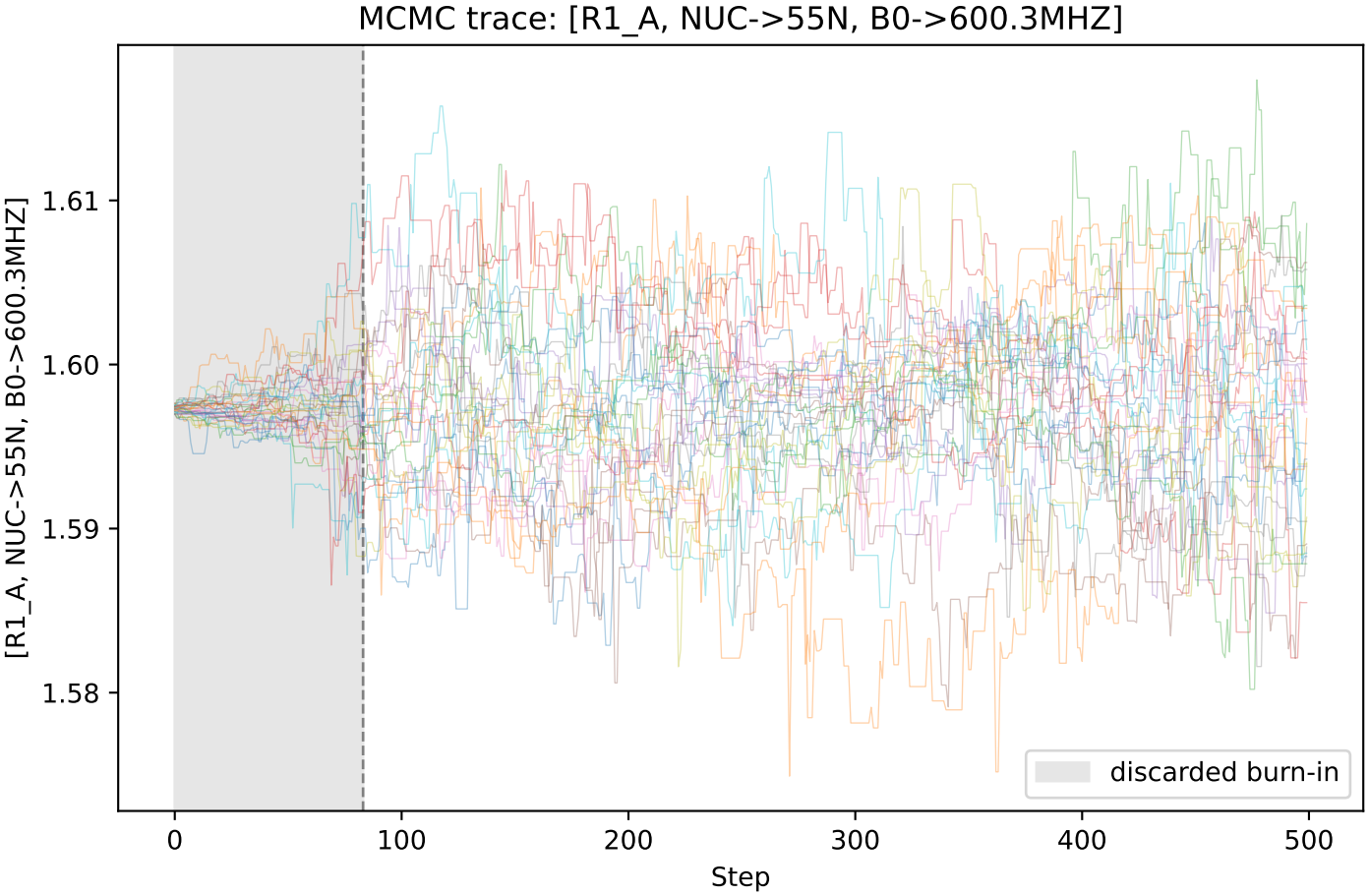

MCMC trace: [DW_AB, NUC->55N]
discarded burn-in
4.80
4.75
4.70
4.65
4.60
[DW_AB, NUC->55N]
0
100
200
300
400
500
Step

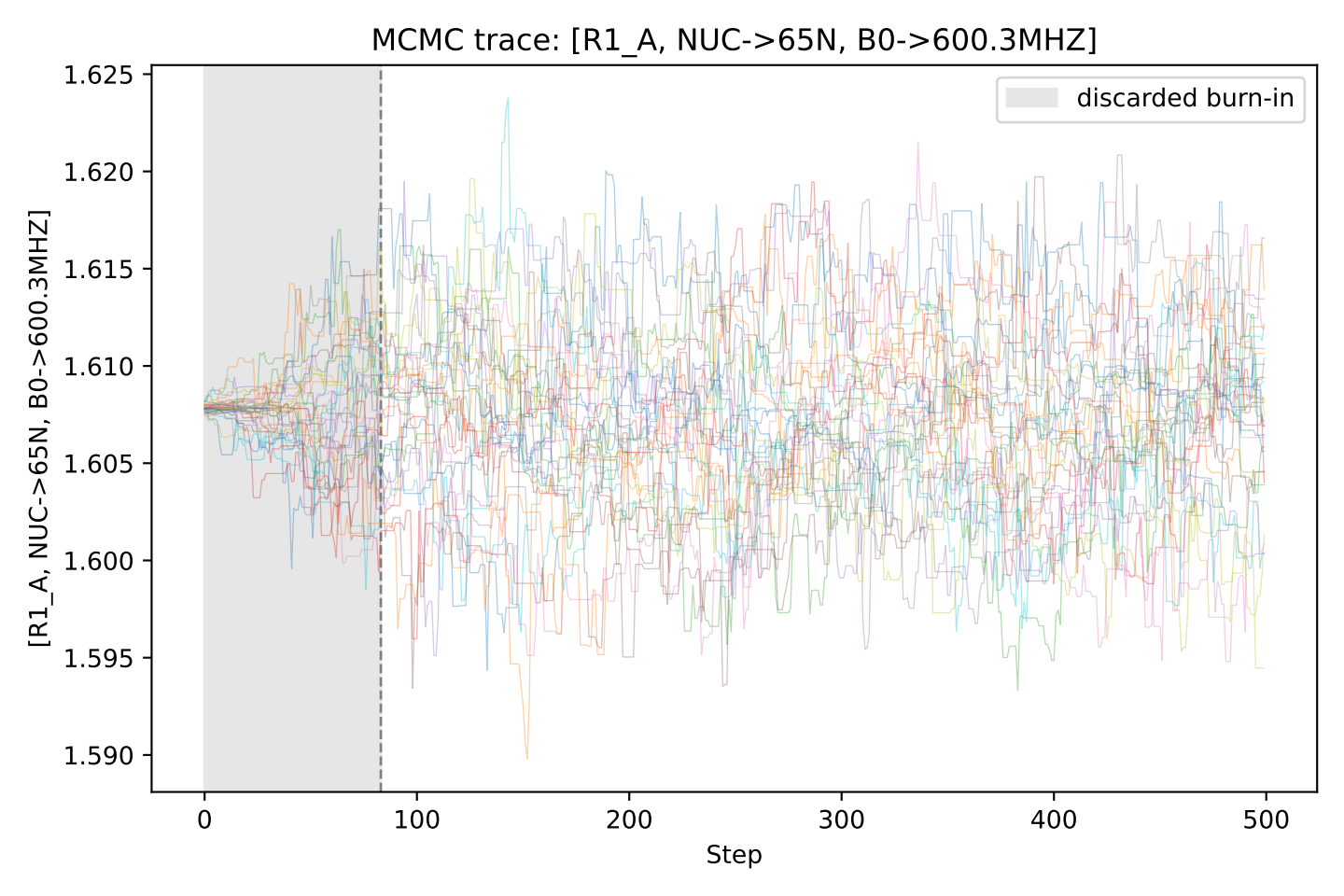

MCMC trace: [DW_AB, NUC->65N]
discarded burn-in
[DW_AB, NUC->65N]
4.15
4.10
4.05
4.00
0
100
200
300
400
500
Step

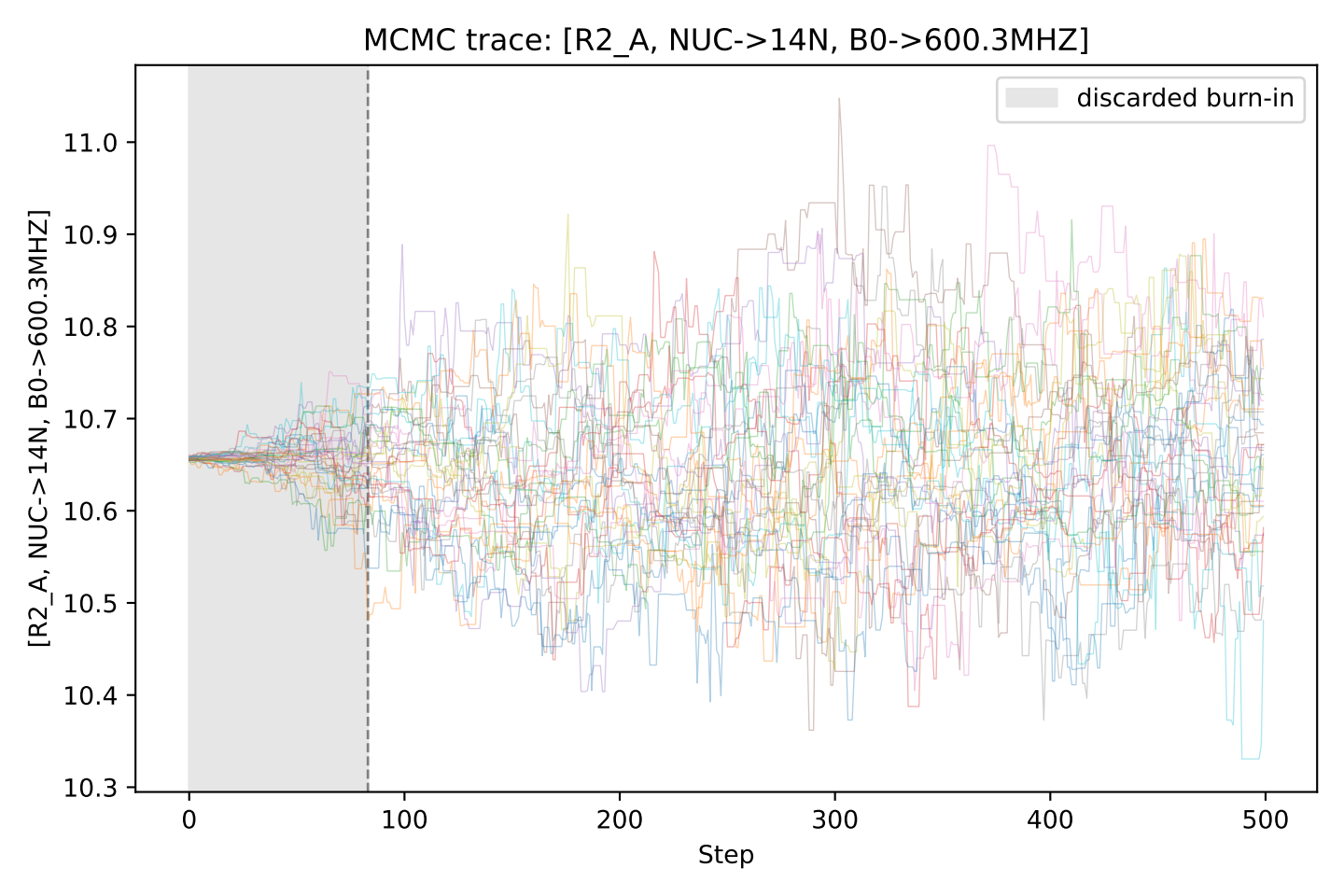

MCMC trace: [KEX_AB]
460
440
420
400
380
[KEX_AB]
0
100
200
300
400
500
Step
discarded burn-in

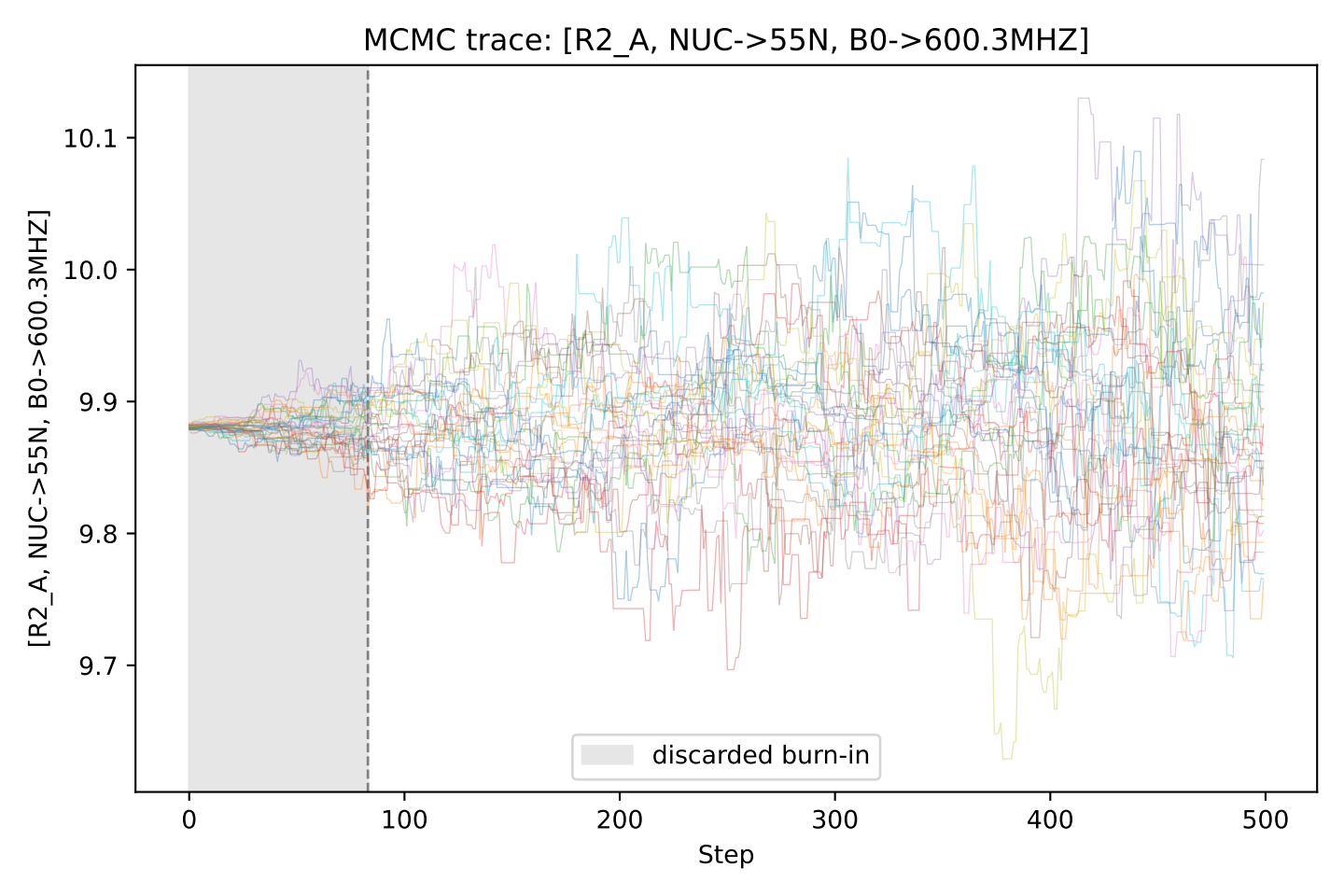

MCMC trace: [PB]
discarded burn-in
0.0037
0.0036
0.0035
0.0034
[PB]
0
100
200
300
400
500
Step

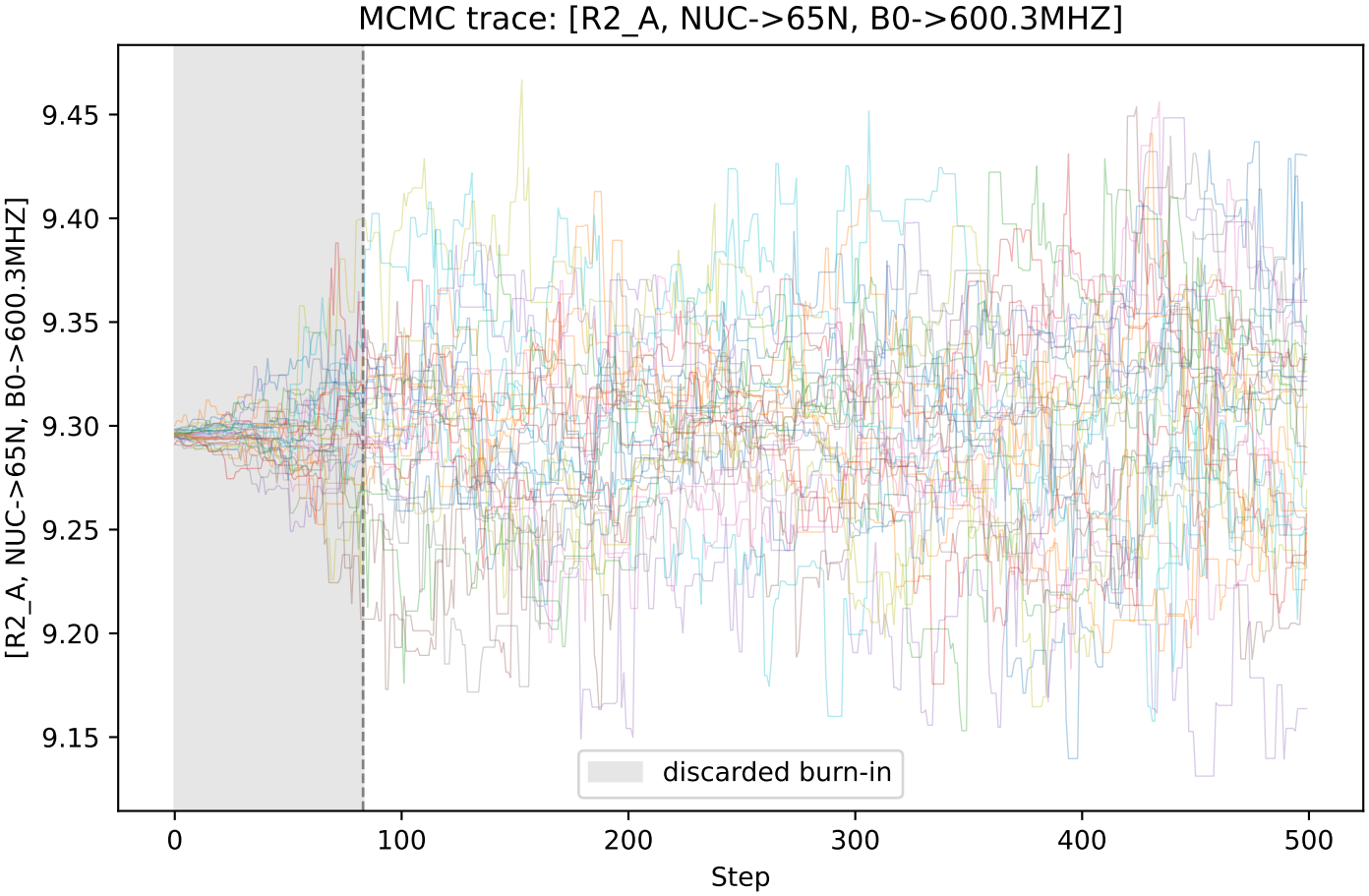

MCMC trace: [R1_A, NUC->14N, B0->600.3MHZ]
discarded burn-in
[R1_A, NUC->14N, B0->600.3MHZ]
1.99
1.98
1.97
1.96
1.95
1.94
0
100
200
300
400
500
Step

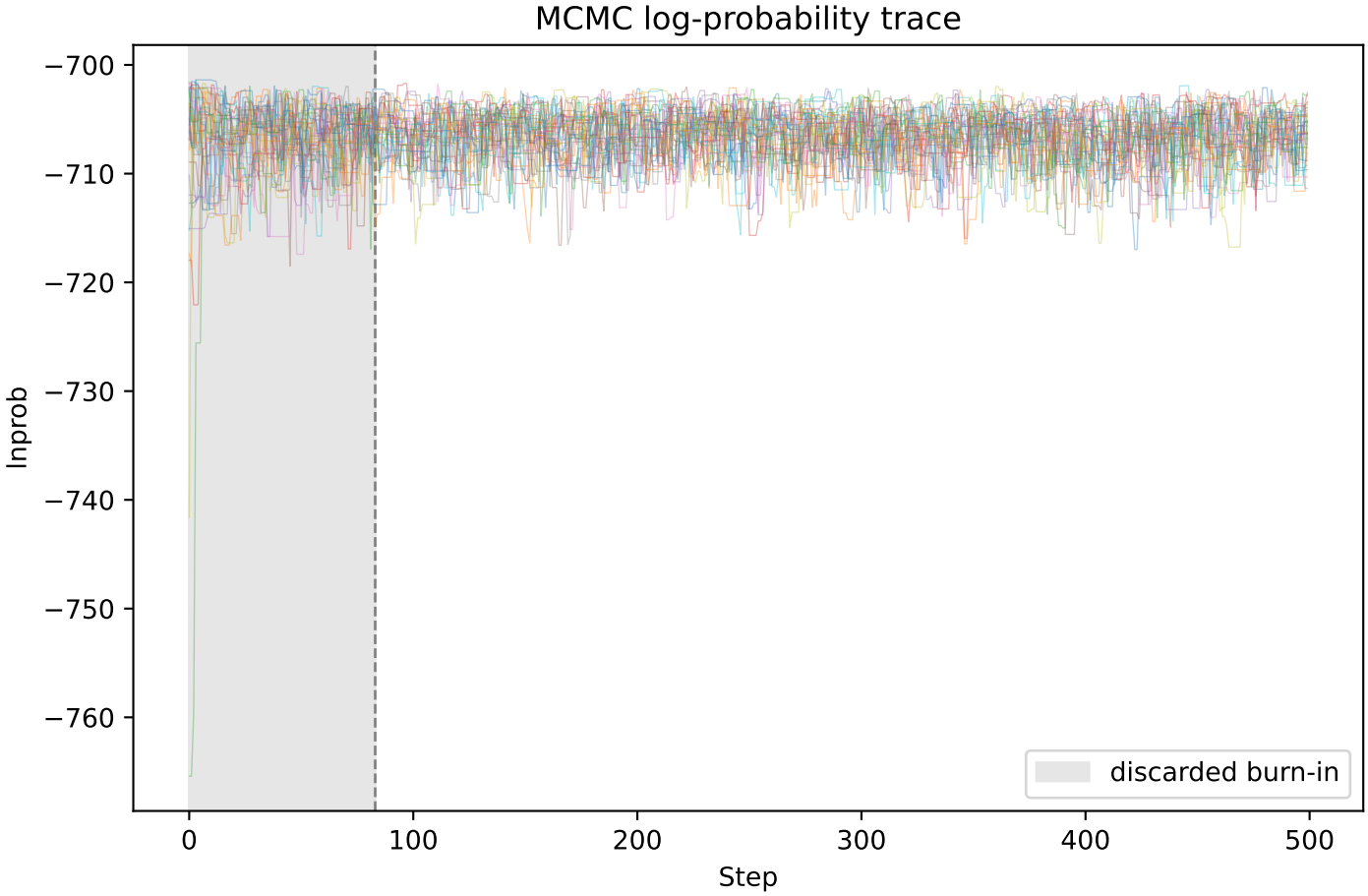

MCMC trace: [R1_A, NUC->55N, B0->600.3MHZ]
[R1_A, NUC->55N, B0->600.3MHZ]
1.61
1.60
1.59
1.58
0
100
200
300
400
500
Step
discarded burn-in

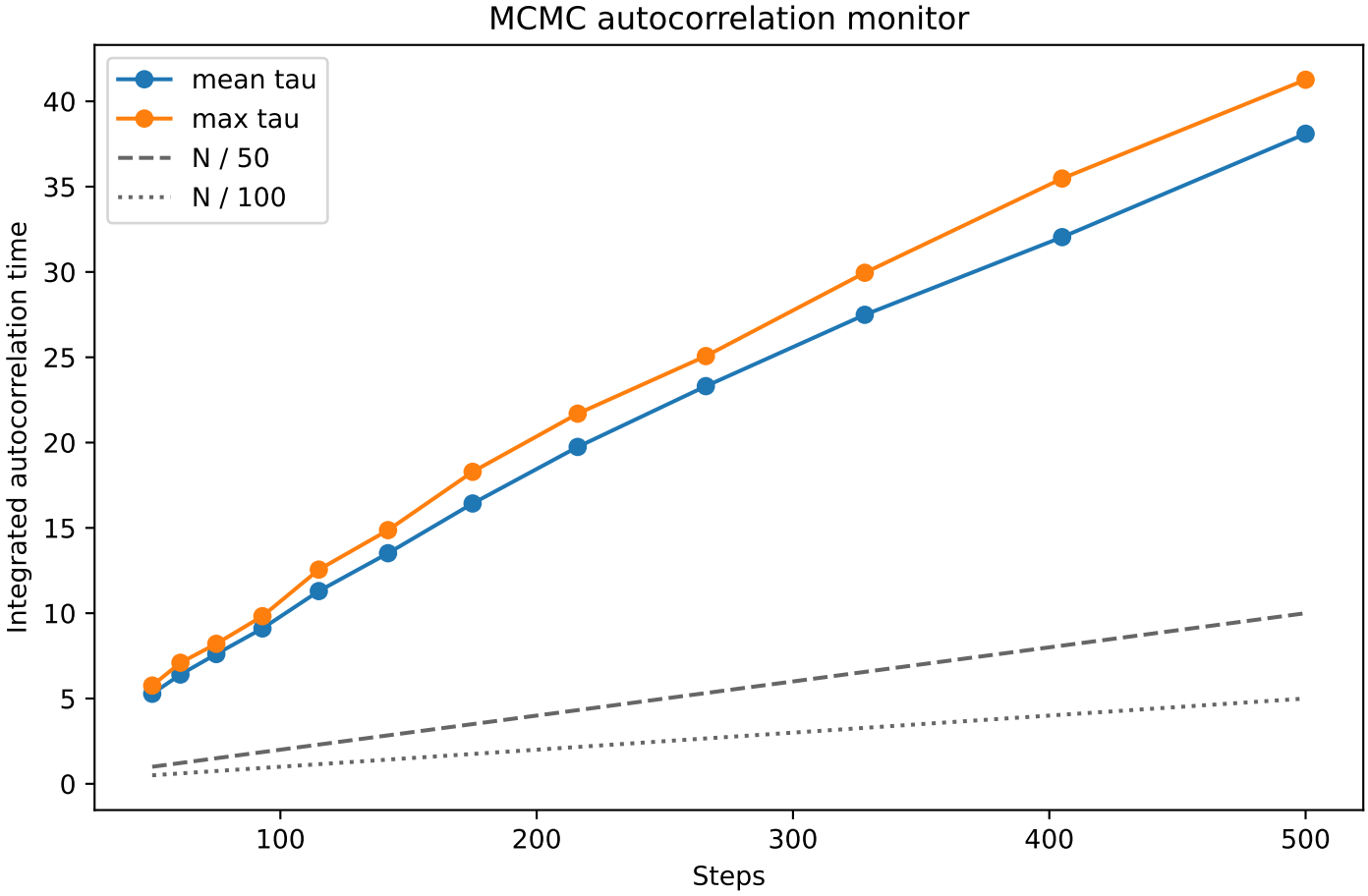

MCMC trace: [R1_A, NUC->65N, B0->600.3MHZ]
discarded burn-in
[R1_A, NUC->65N, B0->600.3MHZ]
1.625
1.620
1.615
1.610
1.605
1.600
1.595
1.590
0
100
200
300
400
500
Step


MCMC trace: [R2_A, NUC->14N, B0->600.3MHZ]
discarded burn-in
[R2_A, NUC->14N, B0->600.3MHZ]
11.0
10.9
10.8
10.7
10.6
10.5
10.4
10.3
0
100
200
300
400
500
Step


MCMC trace: [R2_A, NUC->55N, B0->600.3MHZ]
[R2_A, NUC->55N, B0->600.3MHZ]
10.1
10.0
9.9
9.8
9.7
0
100
200
300
400
500
Step
discarded burn-in


MCMC trace: [R2_A, NUC->65N, B0->600.3MHZ]
[R2_A, NUC->65N, B0->600.3MHZ]
9.45
9.40
9.35
9.30
9.25
9.20
9.15
discarded burn-in
0
100
200
300
400
500
Step


MCMC log-probability trace
lnprob
−700
−710
−720
−730
−740
−750
−760
0
100
200
300
400
500
Step
discarded burn-in


MCMC autocorrelation monitor
mean tau
max tau
N / 50
N / 100
Integrated autocorrelation time
Steps
0
5
10
15
20
25
30
35
40
100
200
300
400
500